Andreas Hovaldt **HØJRUP**

MSc in Computer Engineering, specialized in AI, vision and sound
BSc in Robotics

dreez.dk | linkedin.com/in/andreas-hovaldt | github.com/andreasHovaldt
+45 2135 3504 | andreas.hovaldt@gmail.com
Stokholmvej 29, 9870 Sindal, Denmark



I recently graduated as a robotics and computer engineer, specialized in AI. I finished my MSc in Computer Engineering in June 2026, with a previous bachelor's in robotics from 2024. My work and university experience touched upon a broad range of the ML stack, from applying visual foundation models to LiDAR and camera perception in my AGCO master's thesis, to approximating model predictive controllers during a five-month internship at CERN, down to the GPU-cluster training, experiment tracking, version control using git, and the Linux infrastructure underneath. I enjoy working alongside a team and discussing how to solve problems most efficiently, and when a solution needs implementing, I also like working independently. My native language is Danish, but throughout my work and studies, I have also become fluent in English. I am passionate about modern technology and building things for real world problems. I am now seeking robotics, ML engineering, computer vision research, or data science roles where I can contribute to exciting projects and keep learning along the way. Currently I am situated temporarily in Sindal, but I am planning on relocating for the job.

Work Experience

Jul 2025 - Aug 2025
Feb 2026 - Jun 2026

Student Worker | Applied Vision Research, AGCO A/S, Randers, Denmark

- Researched and evaluated point-cloud segmentation techniques for agricultural environments, laying groundwork for the thesis collaboration
- Trained lightweight task adapters on frozen DINOv3 embeddings to predict machine and material state from in-machine cameras
- Designed a per-instance gating loss that lifted task accuracy from $\sim 70\%$ to over 99%
- Implemented sequence models which beat single-frame (R^2 $0.31 \rightarrow 0.63$)
- Built gradient-rollout explainability tooling

Point Cloud Segmentation | Open3D | PyTorch | DINOv3 | Vision Foundation Models | Weights & Biases | AI Explainability

Sep 2025 - Jan 2026

Computer Engineering Intern, CERN, Geneva, Switzerland

- Built neural-network approximation of an MPC for HVAC air-handling units, cutting online compute by $\sim 460\times$
- Developed the full pipeline : Apache Spark data extraction, dataset curation, modular PyTorch training, and closed-loop simulation deployment
- Compared MLP, TCN, and transformer models, best reached $R^2 = 0.995$ (regression) and 94% class-averaged accuracy (classification)
- Held zone temperatures within operational bounds in closed-loop run

PyTorch | ONNX Runtime | Transformer | Time series | Apache Spark (NXCALS) | Weights & Biases | EcosimPro

Education

2024-2026 **MSc in Computer Engineering** with a specialization in AI, vision, and sound at Aalborg University
Overall average : 11.3 Thesis : 12 (Danish 7-point scale)

2021-2024 **BSc in Robotics** at Aalborg University
Overall average : 10.0 Thesis : 10 (Danish 7-point scale)

Publication

Synthetic LiDAR Data Generation using 3D Gaussian Splatting for 3D Semantic Segmentation Jun 2025
Christiansen, A.T.[*], **Højrup, A.H.**[*], Stephansen, M.K.[*], *et al.* ([*]Equal contribution)
Synthetic Data for Computer Vision Workshop, CVPR
arXiv:2506.05009 | OpenReview

Projects

Foundation Models in Agricultural 3D Perception (MSc Thesis, with AGCO) Feb 2026 - Jun 2026

Evaluated 2D and 3D visual foundation models for in-field LiDAR detection and segmentation, on a self-captured, manually annotated agricultural LiDAR dataset alongside ScanNet v2. Domain fine-tuning a 3D point-cloud foundation model raised detection mAP25 by $\sim 4\times$, while injecting frozen DINOv3 image embeddings lifted trained from-scratch models by $\sim 8.3\times$.

PyTorch | 3DETR | DINOv3 | LiDAR | Pointcept | Slurm | Transformer | Detection | Segmentation

Neural Network Approximation of an MPC for HVAC (CERN) Sep 2025 - Jan 2026

Built neural-network approximations of an economic Model Predictive Controller for HVAC air-handling units at CERN's SR1 facility, with a full pipeline from NXCALS/Spark data extraction to closed-loop deployment against an EcosimPro digital twin. The best (transformer) model reached $R^2 = 0.995$ and 94% class-averaged accuracy while cutting online compute by $\sim 460\times$.

PyTorch | Transformers | TCN | Apache Spark | EcosimPro

Personal Sound Zones for Smartphones Feb 2025 - Jun 2025

Deep-learning sound-zone control that forms a private "bright zone" for the phone user and a "dark zone" around them, allowing enhanced phone call privacy in public spaces, using only native smartphone speakers and microphones and trained entirely on synthetic room impulse responses. Reached 32.5 dB average acoustic contrast, approaching analytically optimal filters.

PyTorch | Acoustics | Signal Processing | Deep Learning | Pyroomacoustics

Bimanual Manipulation of two Kuka IIWA 7 robotic arms with an Integrated LLM Sep 2023 - Jan 2024

Natural-language-controlled bimanual system using two Kuka IIWA 7 arms, with YOLOv8 object detection and a GPT-4 agent translating prompts into robot commands. Achieved a 73% task success rate while demonstrating LLM-driven adaptive task execution.

LLM Agents | YOLOv8 | GPT-4 | ROS2 | Robotics | LangChain

Personal Projects

Self-Hosted Homelab

Self-hosted home server running 10+ containerized services (Docker/LXC) on Proxmox VE, including reverse-proxied web apps, automated media management, game servers, and monitoring. Secured with a WireGuard VPN and Cloudflare mutual-TLS, with internal services kept behind authenticated access.

Proxmox VE | Docker | LXC | WireGuard | Cloudflare | Linux

Personal Finance Tracker

Full-stack web app that fetches and categorizes bank transactions through the GoCardless API, with a FastAPI + PostgreSQL backend and a React frontend, containerized with Docker Compose and self-hosted.

FastAPI | PostgreSQL | React | GoCardless API | Docker Compose

Technical Skills

Programming	Python (advanced); MATLAB, C++, C# (basic)
ML & Vision	PyTorch, scikit-learn, ONNX Runtime, OpenCV, Open3D, Pointcept, Weights & Biases, visual foundation models (DINOv3, Utonia), 3D Gaussian Splatting
Data & Compute	NumPy, SciPy, Pandas, Apache Spark, Slurm GPU clusters (HPC)
Infrastructure	Linux, Docker, Proxmox VE, Git, FastAPI, NGINX, WireGuard
Robotics	ROS2, NVIDIA Isaac Sim, sensor integration (LiDAR, RGB-D cameras), trajectory planning

Interests

Outside of work, I still like to sometimes have something hands-on to tinker with. Previously I have worked on small home electronics projects with either an Arduino or a Raspberry Pi. Recently, I have focused on tinkering with my self-hosted Proxmox setup. When I'm not working on something, I like to go for my weekly runs or, in the weekends, go hiking, and a pleasant evening with friends is always a nice way to unwind.

References

Available upon request.